Zekun Wang

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My research interest is optimizing networking system for higher performance and better security. I'm looking for chances to gain insight into industry and frontier technology by proposing, developing and implementing innovative network frameworks powered by edge computing, machine learning, optimization theory, etc.

Education

University of Tokyo (UTokyo)*, Tokyo, Japan* ***09/2022-Present***
Exchange Study (Finish in 03/2023) *Major: Creative Informatics*

- Relevant Coursework: Network optimizations, Parallel and distributed programming
- Honors: JASSO Scholarship

KTH Royal Institute of Technology (KTH)*, Stockholm, Sweden* ***09/2021-Present***
Master of Science *Major: Communication Systems* *GPA: 4.81 / 5.0*

- Relevant Coursework: Advanced internetworking, Internet security and privacy, SDN & NFV
- Honors: KTH Scholarship (35 / 2200)

Huazhong University of Science and Technology (HUST)*, Wuhan, China* ***09/2018-06/2022***
Bachelor of Engineering *Major: Electronic and Information Engineering* *GPA: 3.89 / 4.0*

- Relevant Coursework: Stochastic processes, Computer networks, Principles and applications of big data technology
- Honors: Outstanding Undergraduates in Term of Academic Performance (awarded to top 1%), Scholarship for Academic Excellence, Excellent Student Association Cadre

Research Experience

Research on Applying Multi-access Edge Computing & Federated Learning in IoV Scenario *, UTokyo* ***10/2022-Present***
Research Project of Tsukada Lab *supervised by Prof. Manabu Tsukada*

- Proposed an IoV framework utilizing MEC to realize Federated Learning among vehicles, caching multiple models in edge nodes.
- Conducted a literature review on MEC and FL used for Vehicle Networking, especially focusing on resource allocation and communication.
- Expected to combine MEC and FL to build a system reducing power consumption while increasing efficiency by model caching and task offloading.

ML-Stealer: Stealing Prediction Functionality of Machine Learning Models*, HUST* ***04/2020-05/2021***
National Undergraduate Training Program for Innovation *supervised by Prof. Chen Wang*

- Implemented Generative Adversarial Network(GAN) to learn the prediction behavior of victim ML models. Improved sampling and data augmentation by Variational Autoencoder(VAE).
- Designed loss function Applying focal-loss strategy to relieve the effect of unevenly distributed training dataset.
- Achieved an averaged accuracy of 85.6% and up to 93.6% at best under the mere black-box access scenario.

Project Experience

Summer Internship in Ericsson*, Kista, Sweden* ***06/2022-09/2022***
Analyzing and Visualizing the Performance of Distributed Bazel Compiling System

- Developed the Bazel Event Service Application in Java, which communicates with multiple Bazel clients, processes and sends the received data to the OpenSearch backend. Use OpenSearch for data storing, large data analysis and visualization.
- Modified the source code of Bazel to enhance the Bazel Event Protocol, making it possible to gain comprehensive client information.
- Followed an Agile workflow, and implemented many cooperative developing methods to increase the team efficiency.

ACME Secure Network Design *, KTH Course Project* ***02/2022-03/2022***
Secure Network System Allowing Multiple Authentication Methods

- Designed the system framework based on FreeIPA, combining OpenVPN, WireGuard, FreeRadius for secure communication, and NextCloud for file-sharing service. Connected two bases by WireGuard while allowing connection from outside by OpenVPN.
- Tested on real devices, where OpenWRT Routers performed as Gateways and private PCs performed as internal servers.

Skills

Programming	Python, C, Java, Matlab, Bash Script, Javascript
Language	English (Professional working proficiency), Chinese (Native speaker)